Mark Scheme

Q1.

Part	Working or answer an examiner might expect to see	Mark	Notes
(a)	$\log a - \log b = \log \frac{a}{b}$	B1	This mark is given for restating the log equation using $\log \frac{a}{b}$
	$a - b = \frac{a}{b}$ $ab - b^2 = a$ $ab - a = b^2$	M1	This mark is given for rearranging so that terms in a are on one side of the equation
	$a(b - 1) = b^2$ $a = \frac{b^2}{(b-1)}$	A1	This mark is for rearranging to show the result required
(b)	$b \neq 1$	B1	This mark is given for deducing that $b \neq 1$
	Since $a > 0$, $\frac{b^2}{(b-1)} > 0$ $b > 1$ since b^2 is positive	B1	This mark is given for stating that $b > 1$ and explaining the reason for the restriction

Q2.

Question	Scheme	Marks	AOs
(a)	$A = 1000$	B1	3.4
	$2000 = 1000e^{5k}$ or $e^{5k} = 2$	M1	1.1b
	$e^{5k} = 2 \Rightarrow 5k = \ln 2 \Rightarrow k = ...$	M1	2.1
	$N = 1000e^{\left(\frac{1}{5}\ln 2\right)t}$ or $N = 1000e^{0.139t}$	A1	3.3
		(4)	
(b)	$\frac{dN}{dt} = 1000 \times \left(\frac{1}{5}\ln 2\right)e^{\left(\frac{1}{5}\ln 2\right)t}$ or $\frac{dN}{dt} = 1000 \times 0.139e^{0.139t}$ $\left(\frac{dN}{dt}\right)_{t=8} = 1000 \times \left(\frac{1}{5}\ln 2\right)e^{8\times\frac{1}{5}\ln 2}$ or $\left(\frac{dN}{dt}\right)_{t=8} = 1000 \times 0.139e^{0.139\times 8}$	M1	3.1b
	= awrt 420	A1	1.1b
		(2)	
(c)	$500e^{1.4\times\left(\frac{1}{5}\ln 2\right)T} = 1000e^{\left(\frac{1}{5}\ln 2\right)T}$ or $500e^{1.4\times"0.139"t} = 1000e^{"0.139"t}$	M1	3.4
	Correct method of getting a linear equation in T E.g. $0.08T\ln 2 = \ln 2$ or $1.4\times"0.339"T = \ln 2 + "0.339"t$	M1	2.1
	$T = 12.5$ hours	A1	1.1b
		(3)	
			(9 marks)
	Notes		

Mark as one complete question. Marks in (a) can be awarded from (b)

(a)

B1: Correct value of A for the model. Award if equation for model is of the form $N = 1000e^{\dots t}$

M1: Uses the model to set up a correct equation in k. Award for substituting $N = 2000, t = 5$ following through on their value for A.

M1: Uses correct ln work to solve an equation of the form $ae^{5k} = b$ and obtain a value for k

A1: Correct equation of model. Condone an ambiguous $N = 1000e^{\frac{1}{5}\ln 2t}$ unless followed by something incorrect. Watch for $N = 1000 \times 2^{\frac{1}{5}t}$ which is also correct

(b)

M1: Differentiates αe^{kt} to βe^{kt} and substitutes $t = 8$ (Condone $\alpha = \beta$ so long as you can see an attempt to differentiate)

A1: For awrt 420 (2sf).

(c)

M1: Uses both models to set up an equation in T using their value for k, but also allow in terms of k

M1: Uses correct processing using lns to obtain a linear equation in T (or t)

A1: Awrt 12.5

Answers to (b) and (c) appearing without working (i.e. from a calculator).
It is important that candidates show sufficient working to make their methods clear.

(b) If candidate has for example $N = 1000e^{0.139t}$, and then writes at $t = 8$ $\frac{dN}{dt} =$ awrt 420 award both marks. Just the answer from a correct model equation score SC 1,0.

(c) The first M1 should be seen E.g $500e^{1.4 \times "0.139"t} = 1000e^{"0.139"t}$

If the answer $T = 12.5$ appears without any further working score SC M1 M1 A0

Q3.

Question	Scheme	Marks	AOs
(a)	(i) Method to find p Eg. Divides $32000 = Ap^4$ by $50000 = Ap^{11}$ $p^7 = \frac{50000}{32000} \Rightarrow p = \sqrt[7]{\frac{50000}{32000}} = ...$	M1	3.1a
	$p = 1.0658$	A1	1.1b
	(ii) Substitutes their $p = 1.0658$ into either equation and finds A $A = \frac{32000}{'1.0658'^4}$ or $A = \frac{50000}{'1.0658'^{11}}$	M1	1.1b
	$A = 24795 \to 24805 \simeq 24\,800$ *	A1*	1.1b
		(4)	
(b)	A / (£) $24\,800$ is the value of the car on 1st January 2001	B1	3.4
	p / 1.0658 is the factor by which the value rises each year. Accept that the value rises by 6.6 % a year (ft on their p)	B1	3.4
		(2)	
(c)	Attempts $100000 = '24800' \times '1.0658'^t$		
	$'1.0658'^t = \frac{100000}{24800}$	M1	3.4
	$t = \log_{1.0658}\left(\frac{100000}{24800}\right)$	dM1	1.1b
	$t = 21.8$ or 21.9	A1	1.1b
	cso 2022	A1	3.2a
		(4)	
			(10 marks)

(a) (i)

M1: Attempts to use both pieces of information within $V = Ap^t$, eliminates A correctly and solves an equation of the form $p^n = k$ to reach a value for p.

Allow for slips on the 32 000 and 50 000 and the values of t.

A1: $p =$ awrt 1.0658

Both marks can be awarded from incorrect but consistent interpretations of t. Eg. $32000 = Ap^5$, $50000 = Ap^{12}$

(a)(ii)

M1: Substitutes their $p = 1.0658$ into either of their equations and finds A

Eg $A = \frac{32000}{1.0658^4}$ or $A = \frac{50000}{1.0658^7}$ but you may follow through on incorrect equations from part (i)

A1*: Shows that A is between 24 795 and 24 805 before you see ' =24 800' or '≈ 24800'. Accept with or without units.

An alternative to (ii) is to start with the given answer.

M1: Attempts $24800 \times '1.0658'^4 = (32000.34)$

A1: 24800×1.0658^4, achieves a value between 31095 and 32005 followed by $\approx 32\,000$ hence A must be $\approx 24\,800$

(b)

B1: States that A is the value of the car on 1st January 2001.
The statement must reference **the car**, its **cost/value**, and **"0" time**
Allow 'it is the initial value of the car" "it is the cost of the car at $t = 0$" "it is the cars starting value"

B1: States that p is the rate at which the value of the car rises each year.
The statement must reference **a yearly rate** and **an increase in value or multiplier**.
They could reference the 1.0658 Eg "The cars value rises by 6.5 % each year."
Allow "p is the rate the cars value is rising each year" "it is the proportional increase in value of the car each year" "the factor by which the value of the car is rising each year" 'its value appreciates by 6.5% per year' Allow ' the value of the car multiplies by p each year'
Do not allow "by how much the value of the car rises each year " or "it is the rate of inflation"

(c)

M1: Uses the model $100000 = '24800' \times '1.0658'^t$ and proceeds to their $'1.0658'^t = k$
Allow use of any inequality here.

dM1: For the complete method of (i) using the information given with their equation of the model **and** (ii) translating the situation into a correct method to find 't'

A1: $(t) =$ awrt 21.8 or 21.9 or $\log_{1.0658}\left(\frac{100000}{24800}\right)$ oe

A1: States in the year 2022. A candidate using a GP formula can be awarded full marks
Allow different methods in part (c).
Eg Via GP a formula
M1: $24800\times'1.0658'^{n-1} = 100000 \Rightarrow '1.0658'^{n-1} = K$
dM1: Uses a correct method to find n.
A2: 2022
Via (trial and improvement)

M1: Uses the model by substituting integer values of t into their $V = Ap^t$ so that for $t = n, V < 100\,000$ or $t = n+1, V > 100\,000$
(So for the correct A and p this would be scored for $t = 21, V \approx £95000$ or $t = 21, V \approx £101\,000$

dM1: For a complete method showing that this is the least value. So both of the above values

A1: Allow for 22 following correct and accurate results (awrt nearest £1000 is sufficient accuracy)

A1: As before

Q4.

Part	**Working or answer an examiner might expect to see**	**Mark**	**Notes**
(a)	The asymptote is found where $2x - q = 0$ Hence $q = 4$	B1	This mark is given for explaining that the asymptote at $x = 2$ is a solution of $2x - q = 0$
	$y = \frac{p-3x}{(2x-4)(x+3)}$ $\frac{1}{2} = \frac{p-9}{(6-4)(3+3)}$	M1	This mark is given for substituting $x = 3$, $y = \frac{1}{2}$ (and $q = 4$)

(b)	$\frac{15-3x}{(2x-4)(x+3)} = \frac{A}{(2x-4)} + \frac{B}{(x+3)}$	M1	This mark is given for a method to use partial fractions
	$= \frac{1.8}{(2x-4)} - \frac{2.4}{(x+3)}$	M1	This mark is given for finding values for A and B
	$= \frac{0.9}{(x-2)} - \frac{2.4}{(x+3)}$	A1	This mark is given for a fully simplified expression
	$I = \int \frac{15-3x}{(2x-4)(x+3)}\,dx$ $= m\ln(2x-4) + n\ln(x+3)$	M1	This mark is given for a method to integrate to find the area of R
	$= 0.9\ln(2x-4) + 2.4\ln(x+3)$	A1	This mark is given for a correct expression for the area of R
	Area $R = \left[0.9\ln(2x-4) - 2.4\ln(x+3)\right]_3^5$	M1	This mark is given for deducing an expression for the area of R ($y = 0$ when $x = 5$)
	$= [0.9\ln 6 - 2.4\ln 8] - [0.9\ln 2 - 2.4\ln 6]$ $= [0.9\ln 6 + 2.4\ln 6] - [7.2\ln 2 + 0.9\ln 2]$ $= 3.3\ln 6 - 8.1\ln 2$ $= 3.3\ln 3 + 3.3\ln 2 - 8.1\ln 2$	M1	This mark is given for a method to find the exact area of R
	$= 3.3\ln 3 - 4.8\ln 2$	A1	This mark is given for a correct value of the area of R with $a = 3.3$ and $b = 4.8$

Q5.

Question	Scheme	Marks	AOs
	Writes $\int \frac{t+1}{t}\,\mathrm{d}t = \int 1+\frac{1}{t}\,\mathrm{d}t$ and attempts to integrate	M1	2.1
	$= t + \ln t \; (+c)$	M1	1.1b
	$(2a+\ln 2a)-(a+\ln a)=\ln 7$	M1	1.1b
	$a=\ln\frac{7}{2}$ with $k=\frac{7}{2}$	A1	1.1b
		(4 marks)	

Notes:

M1: Attempts to divide each term by t or alternatively multiply each term by t^{-1}

M1: Integrates each term and knows $\int \frac{1}{t}\,\mathrm{d}t = \ln t$. The $+\,c$ is not required for this mark

M1: Substitutes in both limits, subtracts and sets equal to ln7

A1: Proceeds to $a=\ln\frac{7}{2}$ and states $k=\frac{7}{2}$ or exact equivalent such as 3.5

Q6.

Question	Scheme	Marks	AOs
(a)	Sets $\dfrac{1}{P(11-2P)} = \dfrac{A}{P} + \dfrac{B}{(11-2P)}$	B1	1.1a
	Substitutes either $P = 0$ or $P = \dfrac{11}{2}$ into $1 = A(11-2P) + BP \Rightarrow A \text{ or } B$	M1	1.1b
	$\dfrac{1}{P(11-2P)} = \dfrac{1/11}{P} + \dfrac{2/11}{(11-2P)}$	A1	1.1b
		(3)	
(b)	Separates the variables $\displaystyle\int \frac{22}{P(11-2P)}\,\mathrm{d}P = \int 1\,\mathrm{d}t$	B1	3.1a
	Uses (a) and attempts to integrate $\displaystyle\int \frac{2}{P} + \frac{4}{(11-2P)}\,\mathrm{d}P = t + c$	M1	1.1b
	$2\ln P - 2\ln(11-2P) = t + c$	A1	1.1b
	Substitutes $t = 0, P = 1 \Rightarrow t = 0, P = 1 \Rightarrow c = (-2\ln 9)$	M1	3.1a
	Substitutes $P = 2 \Rightarrow t = 2\ln 2 + 2\ln 9 - 2\ln 7$	M1	3.1a
	Time $= 1.89$ years	A1	3.2a
		(6)	

(c)	Uses ln laws $2\ln P - 2\ln(11-2P) = t - 2\ln 9$ $\Rightarrow \ln\left(\dfrac{9P}{11-2P}\right) = \dfrac{1}{2}t$	M1	2.1
	Makes 'P' the subject $\Rightarrow \left(\dfrac{9P}{11-2P}\right) = e^{\frac{1}{2}t}$ $\Rightarrow 9P = (11-2P)e^{\frac{1}{2}t}$ $\Rightarrow P = f\left(e^{\frac{1}{2}t}\right)$ or $\Rightarrow P = f\left(e^{-\frac{1}{2}t}\right)$	M1	2.1
	$\Rightarrow P = \dfrac{11}{2 + 9e^{-\frac{1}{2}t}} \Rightarrow A = 11, B = 2, C = 9$	A1	1.1b
		(3)	
			(12 marks)

Notes:

(a)

B1: Sets $\dfrac{1}{P(11-2P)}=\dfrac{A}{P}+\dfrac{B}{(11-2P)}$

M1: Substitutes $P=0$ or $P=\dfrac{11}{2}$ into $1=A(11-2P)+BP\Rightarrow A \text{ or } B$

Alternatively compares terms to set up and solve two simultaneous equations in A and B

A1: $\dfrac{1}{P(11-2P)}=\dfrac{1/11}{P}+\dfrac{2/11}{(11-2P)}$ or equivalent $\dfrac{1}{P(11-2P)}=\dfrac{1}{11P}+\dfrac{2}{11(11-2P)}$

Note: The correct answer with no working scores all three marks.

(b)

B1: Separates the variables to reach $\displaystyle\int\frac{22}{P(11-2P)}\mathrm{d}P=\int 1\mathrm{d}t$ or equivalent

M1: Uses part (a) and $\displaystyle\int\frac{A}{P}+\frac{B}{(11-2P)}\mathrm{d}P=A\ln P\pm C\ln(11-2P)$

A1: Integrates both sides to form a correct equation including a 'c' Eg $2\ln P-2\ln(11-2P)=t+c$

M1: Substitutes $t=0$ and $P=1$ to find c

M1: Substitutes $P=2$ to find t. This is dependent upon having scored both previous M's

A1: Time = 1.89 years

(c)

M1: Uses correct log laws to move from $2\ln P-2\ln(11-2P)=t+c$ to $\ln\left(\dfrac{P}{11-2P}\right)=\dfrac{1}{2}t+d$ for their numerical 'c'

M1: Uses a correct method to get P in terms of $e^{\frac{1}{2}t}$

This can be achieved from $\ln\left(\dfrac{P}{11-2P}\right)=\dfrac{1}{2}t+d\Rightarrow\dfrac{P}{11-2P}=e^{\frac{1}{2}t+d}$ followed by cross multiplication and collection of terms in P (See scheme)

Alternatively uses a correct method to get P in terms of $e^{-\frac{1}{2}t}$ For example $\dfrac{P}{11-2P}=e^{\frac{1}{2}t+d}\Rightarrow\dfrac{11-2P}{P}=e^{-\left(\frac{1}{2}t+d\right)}\Rightarrow\dfrac{11}{P}-2=e^{-\left(\frac{1}{2}t+d\right)}\Rightarrow\dfrac{11}{P}=2+e^{-\left(\frac{1}{2}t+d\right)}$ followed by division

A1: Achieves the correct answer in the form required. $P=\dfrac{11}{2+9e^{-\frac{1}{2}t}}\Rightarrow A=11, B=2, C=9$ oe

Q7.

Question	Scheme	Marks	AOs
	Any equation involving an exponential of the correct form. See notes	M1	3.1b
	$n = Ae^{kt}$ (where A and k are positive constants)	A1	1.1b
		(2)	
			(2 marks)
Notes:			

M1: Any equation of the correct form, involving n and an exponential in t.

So allow for example $n = e^{\pm t}$, $n = Ae^{\pm t}$, $n = Ae^{\pm kt}$ condoning $n = A + Be^{\pm t}$

Condone an intermediate form where n has not been made the subject.

E.g. Allow $\ln n = kt + c$ but also condone $\ln n = kt$

A1: E.g. $n = Ae^{kt}$, $n = e^{kt+c}$, $n = e^{kt}e^{c}$ There is no requirement to state that A and k are positive constants

Note that the two constants need to be different.

Mark the final answer so $n = e^{kt+c}$ followed by $n = e^{kt} + e^{c}$ o.e. $n = e^{kt} + A$ such as is M1 A0

You may see solutions that don't include "e".

These are fine so you can include versions of $n = Ak^{t}$ using the same marking criteria as above

FYI $\frac{dn}{dt} = Ak^{t} \times \ln k = \ln k \times n$ so $\frac{dn}{dt} \propto n$

Q8.

Question	Scheme	Marks	AOs
	$\log_3(12y+5) - \log_3(1-3y) = 2 \Rightarrow \log_3 \frac{12y+5}{1-3y} = 2$ or e.g. $2 = \log_3 9$	B1 M1 on EPEN	1.1b
	$\log_3 \frac{12y+5}{1-3y} = 2 \Rightarrow \frac{12y+5}{1-3y} = 3^2 \Rightarrow 9 - 27y = 12y + 5 \Rightarrow y = ...$ or e.g. $\log_3(12y+5) = \log_3(3^2(1-3y)) \Rightarrow (12y+5) = 3^2(1-3y) \Rightarrow y = ...$	M1	2.1
	$y = \frac{4}{39}$	A1	1.1b
		(3)	
			(3 marks)

Notes

B1(M1 on EPEN): Applies at least one addition or subtraction law of logs correctly.
Can also be awarded for using $2 = \log_3 9$. This may be implied by e.g. $\log_3 ... = 2 \Rightarrow ... = 9$

M1: A rigorous argument with no incorrect working to remove the log or logs correctly and obtain a correct equation in any form **and** solve for y.

A1: Correct **exact** value. Allow equivalent fractions.

Guidance on how to mark particular cases:

$$\log_3(12y+5)-\log_3(1-3y)=2 \Rightarrow \frac{\log_3(12y+5)}{\log_3(1-3y)}=2$$
$$\Rightarrow \frac{12y+5}{1-3y}=3^2 \Rightarrow 9-27y=12y+5 \Rightarrow y=\frac{4}{39}$$

B1M0A0

$$\log_3(12y+5)-\log_3(1-3y)=2 \Rightarrow \frac{\log_3(12y+5)}{\log_3(1-3y)}=2 \Rightarrow \log_3\frac{12y+5}{1-3y}=2$$
$$\Rightarrow \frac{12y+5}{1-3y}=3^2 \Rightarrow 9-27y=12y+5 \Rightarrow y=\frac{4}{39}$$

B1M0A0

$$\log_3(12y+5)-\log_3(1-3y)=2 \Rightarrow \frac{12y+5}{1-3y}=3^2 \Rightarrow 9-27y=12y+5 \Rightarrow y=\frac{4}{39}$$

B1M1A1

Q9.

Question	Scheme	Marks	AOs
(a)	$T=al^b \Rightarrow \log_{10}T=\log_{10}a+\log_{10}l^b$	M1	2.1
	$\Rightarrow \log_{10}T=\log_{10}a+b\log_{10}l$ * or $\Rightarrow \log_{10}T=b\log_{10}l+\log_{10}a$ *	A1*	1.1b
		(2)	
(b)	$b=0.495$ or $b=\frac{45}{91}$	B1	2.2a
	$0=\text{"}0.495\text{"}\times -0.7+\log_{10}a \Rightarrow a=10^{0.346...}$ or $0.45=\text{"}0.495\text{"}\times 0.21+\log_{10}a \Rightarrow a=10^{0.346...}$	M1	3.1a
	$T=2.22l^{0.495}$	A1	3.3
		(3)	
(c)	The time taken for one swing of a pendulum of length 1 m	B1	3.2a
		(1)	
			(6 marks)

Notes
(a) M1: Takes logs of both sides and shows the addition law. Implied by $T = al^b \Rightarrow \log_{10} a + \log_{10} l^b$ A1*: Uses the power law to obtain the given equation with no errors. Allow the bases to be missing in the working but they must be present in the final answer. Also allow t rather than T and A rather than a. **Allow working backwards e.g.** $\log_{10} T = b\log_{10} l + \log_{10} a \Rightarrow \log_{10} T = \log_{10} l^b + \log_{10} a$ $\Rightarrow \log_{10} T = \log_{10} al^b \Rightarrow T = al^b$ * M1: Uses the given answer and uses the power law and addition law correctly A1: Reaches the given equation with no errors as above (b) B1: Deduces the correct value for b (Allow awrt 0.495 or $\frac{45}{91}$) M1: Correct strategy to find the value of a. E.g. substitutes one of the given points and their value for b into $\log_{10} T = \log_{10} a + b\log_{10} l$ and uses correct log work to identify the value of a. Allow slips in rearranging their equation but must be correct log work to find a. Alternatively finds the equation of the straight line and equates the constant to $\log_{10} a$ and uses correct log work to identify the value of a. E.g. $y - 0.45 = "0.495"(x - 0.21) \Rightarrow y = "0.495"x + 0.346 \Rightarrow a = 10^{0.346} = ...$ A1: Complete equation $T = 2.22l^{0.495}$ or $T = 2.22l^{\frac{45}{91}}$ (Allow awrt 2.22 and awrt 0.495 or $\frac{45}{91}$) **Must see the <u>equation</u> not just correct values as it is a requirement of the question.** (c) B1: Correct interpretation

Q10.

Part	Working or answer an examiner might expect to see	Mark	Notes
(a)	$y = x^x \Rightarrow \ln y = x \ln x$	M1	This mark is for a method to find the x-coordinate of the turning point of C by taking logarithms
	$\ln y = x \ln x \Rightarrow \frac{1}{y}\frac{dy}{dx} = \ln x + 1$	M1	This mark is given for a method using implicit differentiation
		A1	This mark is given for a correct expression for $\frac{1}{y}\frac{dy}{dx}$
	Setting $\frac{dy}{dx} = 0$, $\ln x + 1 = 0$	M1	This mark is given for a method for finding the turning point of C by setting $\frac{dy}{dx} = 0$
	$x = e^{-1}$	A1	This mark is given for correctly finding a value for the x-coordinate of the turning point of C

Part	Working or answer an examiner might expect to see	Mark	Notes
(b)	$1.5^{1.5} = 1.837\ldots$, $1.6^{1.6} = 2.121\ldots$	M1	This mark is given for substituting 1.5 and 1.6 into $y = x^x$
	The curve C contains the points (1.5, 1.8) and (1.6, 2.1). At P, $y = 2$ Since C is continuous, $1.5 < \alpha < 1.6$	A1	This mark is given for a valid explanation that C contains the points (1.5, 1.8) and (1.6, 2.1) and is continuous
(c)	$x_1 = 1.5$ $x_2 = 2 \times 1.5^{-0.5} = 1.633$	M1	This mark is given for finding a correct value for x_2
	$x_3 = 2 \times 1.633^{-0.633} = 1.466$ $x_4 = 2 \times 1.466^{-0.466} = 1.673$	A1	This mark is given for finding a correct value for x_4
(d)	For example: x_n oscillates is periodic is non-convergent	B1	This mark is given for a valid statement about the long-term behaviour of x_n
	between 1 and 2	B1	This mark is given for stating that the behaviour is between 1 and 2
			(Total 11 marks)

EXAM PAPERS PRACTICE

Q11.

Part	Working or answer an examiner might expect to see	Mark	Notes
(i)	$\sum_{r=4}^{\infty} 20 \times \left(\frac{1}{2}\right)^r$ $= \sum_{r=1}^{\infty} 20 \times \left(\frac{1}{2}\right)^r - (10 + 5 + 2.5)$	M1	This mark is given for a method to find the sum to infinity of a GP
	$= \dfrac{10}{1-\frac{1}{2}} - (10 + 5 + 2.5)$	M1	This mark is given for a method to use a correct sum formula with a correct first term
	$= 2.5$	A1	This mark is given for a correct value for the sum
(ii)	$\sum_{n=1}^{48} \log_5 \left(\frac{n+2}{n+1}\right)$ $= \log_5 \frac{50}{49} + \log_5 \frac{49}{48} + \ldots + \log_5 \frac{4}{3} + \log_5 \frac{3}{2}$	M1	This mark is given for writing out at least four terms of the sum, including the first two and the last two
	$= \log_5 \dfrac{3 \times 4 \times \ldots \times 48 \times 49 \times 50}{2 \times 3 \times 4 \times \ldots \times 48 \times 49} = \log_5 \dfrac{50}{2}$	M1	This mark is given for using the rules of logs and cancelling terms
	$= 2$	A1	This mark is given for a full proof to show the expression is equal to 2 as required
			(Total 6 marks)

Q12.

Question	Scheme	Marks	AOs
(a)	$\frac{dH}{dt}=\frac{H\cos 0.25t}{40} \Rightarrow \int \frac{1}{H}\,dH=\int \frac{\cos 0.25t}{40}\,dt$	M1	3.1a
	$\ln H=\frac{1}{10}\sin 0.25t\,(+c)$	M1 A1	1.1b 1.1b
	Substitutes $t=0, H=5 \Rightarrow c=\ln(5)$	dM1	3.4
	$\ln\left(\frac{H}{5}\right)=\frac{1}{10}\sin 0.25t \Rightarrow H=5e^{0.1\sin 0.25t}$ *	A1*	2.1
		(5)	
(b)	Max height = $5e^{0.1}=5.53$ m (Condone lack of units)	B1	3.4
		(1)	
(c)	Sets $0.25t=\frac{5\pi}{2}$	M1	3.1b
	31.4	A1	1.1b
		(2)	
			(8 marks)

(a)

M1: Separates the variables to reach $\int \frac{1}{H}\,dH=\int \frac{\cos 0.25t}{40}\,dt$ or equivalent.

The integral signs need to be present on both sides and the dH AND dt need to be in the correct positions.

M1: Integrates both sides to reach $\ln H=A\sin 0.25t$ or equivalent with or without the $+c$

A1: $\ln H=\frac{1}{10}\sin 0.25t+c$ or equivalent with or without the $+c$. Allow two constants, one either side

If the 40 was on the lhs look for $40\ln H=4\sin 0.25t+c$ or equivalent.

dM1: Substitutes $t=0, H=5 \Rightarrow c=..$ There needs to have been a single "$+c$" to find.

It is dependent upon the previous M mark. You may allow even if you don't explicitly see "$t=0, H=5$" as it may be implied from their previous line

If the candidate has attempted to change the subject and made an error/ slip then condone it for this M but not the final A. Eg. $40\ln H=4\sin 0.25t+c \Rightarrow\Rightarrow H^{40}=e^{4\sin 0.25t}+e^{c} \Rightarrow 5^{40}=1+e^{c} \Rightarrow c=...$

Also many students will be attempting to get to the given answer so condone the method of finding c =...
These students will lose the A1* mark

A1*: Proceeds via $\ln H=\frac{1}{10}\sin 0.25t+\ln 5$ or equivalent to the given answer $H=5e^{0.1\sin 0.25t}$ with at least one correct intermediate line and no incorrect work.

DO NOT condone c's going to c's when they should be e^{c} or A

Accept as a minimum $\ln H = \frac{1}{10}\sin 0.25t + \ln 5 \Rightarrow H = e^{\frac{1}{10}\sin 0.25t + \ln 5}$ or $H = e^{\frac{1}{10}\sin 0.25t} \times e^{+\ln 5}$ before sight of the given answer

If the only error was to omit the integration signs on line 1, thus losing the first M1, allow the candidate to have access to this mark following a correct intermediate line (see above).

If they attempt to change the subject first then the constant of integration must have been adapted if the A1* is to be awarded. $\ln H = \frac{1}{10}\sin 0.25t + c \Rightarrow H = e^{\frac{1}{10}\sin 0.25t + c} \Rightarrow H = Ae^{\frac{1}{10}\sin 0.25t}$

The dM1 and A1* under this method are awarded at virtually the same time.

Also, for the final two marks, you may see a proof from $\int_0^H \frac{40}{H}\,dH = \int_5^t \cos 0.25t\,dt$

There is an alternative via the use of an integrating factor.

(b)

B1: States that the maximum height is 5.53 m Accept $5e^{0.1}$ Condone a lack of units here, but penalise if incorrect units are used.

(c)

M1: For identifying that it would reach the maximum height for the 2nd time when $0.25t = \frac{5\pi}{2}$ or 450

A1: Accept awrt 31.4 or 10π Allow if units are seen

Q13.

Part	Working or answer an examiner might expect to see	Mark	Notes
(a)	If $d = kV^n$, then $\log_{10} d = \log_{10} k + n \log_{10} V$	M1	This mark is given for
	Plotting $\log_{10} d$ against $\log_{10} V$ will result in a straight line with gradient n and intercept $\log_{10} k$	A1	This mark is given for an explanation of why the second graph shows that $d = kV^n$
	$\log_{10} k = -1.77$ $k = 10^{-1.77} = 0.01698\ldots \approx 0.017$	A1	This mark is for showing fully that $k \approx 0.017$
(b)	$d = kV^n$ When $V = 30$, $d = 20$ and $k = 0.17$ then $20 = 0.017 \times 30^n$	M1	This mark is given for substituting in the formula as a method to find the value of n
	$n \log 30 = \log\left(\frac{20}{0.017}\right)$	M1	This mark is given for a correct expression for n
	$n = 2.08$ to 3 significant figures $d = 0.017 \times V^{2.08}$	A1	This mark is given for finding a correct value of n to 3 significant figures and writing a complete equation for the model
(c)	$\frac{60}{3600} \times 0.8 \times 1000 = 13.33$ m	M1	This mark is given for a method to find the distance, in metres, covered in the reaction time of 0.8 seconds
	$d = 0.017 \times 60^{2.08} = 84.92$ m	M1	This mark is given for a method to use the formula to find the stopping distance
	13.33 m + 84.92 m = 98.25 m Sean will be able to stop before reaching the puddle	A1	This mark is given for finding a correct value of the total stopping distance and giving a valid conclusion
			(Total 9 marks)

Q14.

Question	Scheme	Marks	AOs
(a)	$N = aT^b \Rightarrow \log_{10} N = \log_{10} a + \log_{10} T^b$	M1	2.1
	$\Rightarrow \log_{10} N = \log_{10} a + b\log_{10} T$ so $m = b$ **and** $c = \log_{10} a$	A1	1.1b
		(2)	
(b)	Uses the graph to find either a or b $a = 10^{\text{intercept}}$ **or** $b =$ gradient	M1	3.1b
	Uses the graph to find both a and b $a = 10^{\text{intercept}}$ **and** $b =$ gradient	M1	1.1b
	Uses $T = 3$ in $N = aT^b$ with their a and b	M1	3.1b
	Number of microbes ≈ 800	A1	1.1b
		(4)	
(c)	$N = 1000000 \Rightarrow \log_{10} N = 6$	M1	3.4
	We cannot 'extrapolate' the graph and assume that the model still holds	A1	3.5b
		(2)	
(d)	States that 'a' is the number of microbes 1 day after the start of the experiment	B1	3.2a
		(1)	
			(9 marks)

Notes:

(a)

M1: Takes logs of both sides and shows the addition law

M1: Uses the power law, writes $\log_{10} N = \log_{10} a + b\log_{10} T$ and states $m = b$ **and** $c = \log_{10} a$

(b)

M1: Uses the graph to find either a or b $a = 10^{\text{intercept}}$ **or** $b =$ gradient. This would be implied by the sight of $b = 2.3$ **or** $a = 10^{1.8} \approx 63$

M1: Uses the graph to find both a and b $a = 10^{\text{intercept}}$ **and** $b =$ gradient. This would be implied by the sight of $b = 2.3$ **and** $a = 10^{1.8} \approx 63$

M1: Uses $T = 3 \Rightarrow N = aT^b$ with their a and b. This is implied by an attempt at $63 \times 3^{2.3}$

A1: Accept a number of microbes that are approximately 800. Allow 800 ± 150 following correct work.

There is an alternative to this using a graphical approach.

M1: Finds the value of $\log_{10} T$ from T=3. Accept as $T = 3 \Rightarrow \log_{10} T \approx 0.48$

M1: Then using the line of best fit finds the value of $\log_{10} N$ from their "0.48"

Accept $\log_{10} N \approx 2.9$

M1: Finds the value of N from their value of $\log_{10} N$ $\log_{10} N \approx 2.9 \Rightarrow N = 10^{'2.9'}$

A1: Accept a number of microbes that are approximately 800. Allow 800 ± 150 following correct work

(c)	
M1	For using $N = 1000000$ and stating that $\log_{10} N = 6$
A1:	Statement to the effect that "we only have information for values of $\log N$ between 1.8 and 4.5 so we cannot be certain that the relationship still holds". "We cannot extrapolate with any certainty, we could only interpolate" There is an alternative approach that uses the formula.
M1:	Use $N = 1000000$ in their $N = 63 \times T^{2.3} \Rightarrow \log_{10} T = \dfrac{\log_{10}\left(\dfrac{1000000}{63}\right)}{2.3} \approx 1.83$.
A1:	The reason would be similar to the main scheme as we only have $\log_{10} T$ values from 0 to 1.2. We cannot 'extrapolate' the graph and assume that the model still holds
(d)	
B1:	Allow a numerical explanation $T = 1 \Rightarrow N = a1^b \Rightarrow N = a$ giving a is the value of N at $T = 1$

Q15.

Question	Scheme	Marks	AOs
(a)	Uses or implies $h = 0.5$	B1	1.1b
	For correct form of the trapezium rule =	M1	1.1b
	$\frac{0.5}{2}\{3+2.2958+2(2.3041+1.9242+1.9089)\}=4.393$	A1	1.1b
		(3)	
(b)	Any valid statement reason, for example • Increase the number of strips • Decrease the width of the strips • Use more trapezia	B1	2.4
		(1)	
(c)	For integration by parts on $\int x^2 \ln x \, dx$	M1	2.1
	$=\frac{x^3}{3}\ln x-\int\frac{x^2}{3}\,dx$	A1	1.1b
	$\int -2x+5\,dx=-x^2+5x \quad (+c)$	B1	1.1b
	All integration attempted and limits used Area of $S=\int_1^3 \frac{x^2\ln x}{3}-2x+5\,dx=\left[\frac{x^3}{9}\ln x-\frac{x^3}{27}-x^2+5x\right]_{x=1}^{x=3}$	M1	2.1
	Uses correct ln laws, simplifies and writes in required form	M1	2.1
	Area of $S=\frac{28}{27}+\ln 27 \quad (a=28, b=27, c=27)$	A1	1.1b
		(6)	
			(10 marks)

Notes:	
(a)	
B1:	States or uses the strip width $h = 0.5$. This can be implied by the sight of $\frac{0.5}{2}\{...\}$ in the trapezium rule
M1:	For the correct form of the bracket in the trapezium rule. Must be y values rather than x values $\{\text{first } y \text{ value} + \text{last } y \text{ value} + 2\times(\text{sum of other } y \text{ values})\}$
A1:	4.393
(b)	
B1:	See scheme
(c)	
M1:	Uses integration by parts the right way around. Look for $\int x^2 \ln x \, dx = Ax^3 \ln x - \int Bx^2 \, dx$
A1:	$\frac{x^3}{3}\ln x - \int \frac{x^2}{3} \, dx$
B1:	Integrates the $-2x+5$ term correctly $= -x^2 + 5x$
M1:	All integration completed and limits used
M1:	Simplifies using ln law(s) to a form $\frac{a}{b} + \ln c$
A1:	Correct answer only $\frac{28}{27} + \ln 27$